

Challenge 01

The Task:

- Analyze 18 months of Meridian's procurement data to identify the single system bottleneck using the **Theory of Constraints (TOC)** framework.
- Examine vendor invitations, response rates, bids received, clarification rounds, cycle times, and vendor health to distinguish the true constraint from symptoms.
- Create a **constraint map supported by three charts and data evidence**, with particular attention to changes after the late-2024 disruption.
- Deliver a clear one-sentence constraint statement and two evidence-based sentences explaining why it is the system's chokepoint.
- Propose one immediate, **zero-budget exploitation action** to increase throughput, with Steps 3–5 reserved for a future project.

Input:

- RFQ Data either as csv files or stored in HANA Database
- *Briefing.md* that describes your mission to be completed
- *starter.py* that contains the Boilerplate Code

Output:

1. Complete starter.py with necessary code that generates 4 graphs:
2. **Cycle time** answers the question "*Is the RFQ window too short?*"
3. **Invited / Responding / Bidding** answers the question "Are all vendors invited responding to your bid or not ?, Is your vendor pool healthy ?"
4. **Vendor health scores** answers the question "What drove the pool collapse?"
5. **Clarification rounds** answers the question "Is increase in rounds a Symptom or a constraint?"

Solution:

There are many ways to arrive at the solution, in this workshop, we can try out two options:

Joule Desktop

Claude Code

Joule Desktop

Open Joule Desktop -> New Conversation

Select SAP HANA as connector (*ensure that connector is in connected state, if not follow the set up guide.*)

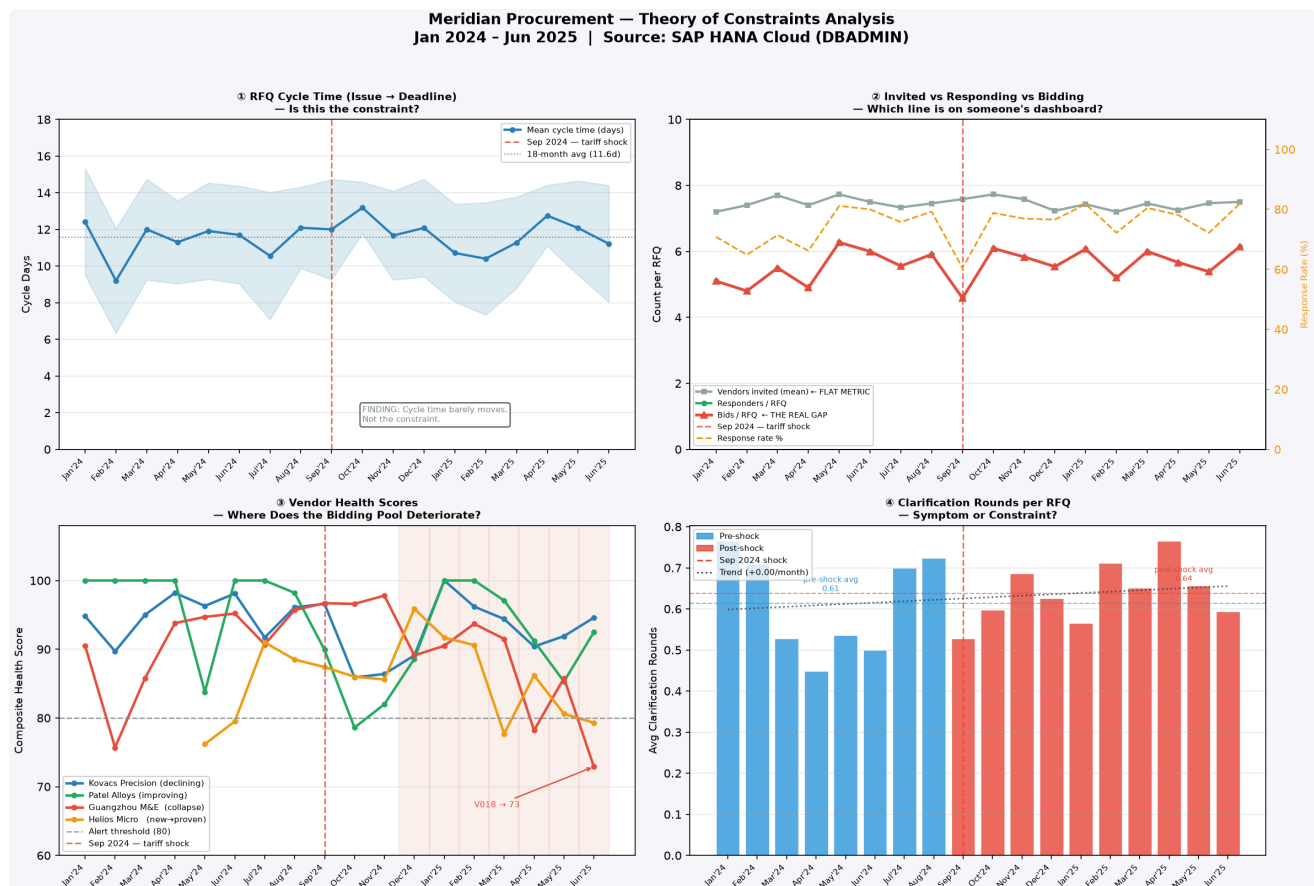
Attach Briefing.md and starter.py from challenges/challenge_01_toc folder to the conversation

Enter the following prompt

Attached Briefing.md contains the challenge. use HANA database to analyse the data instead of CSVs. Complete the starter.py code and run the same

Let it run and complete the task. You should see the following output:

Results



Joule Desktop also gives the reason and solution, please read the same carefully

The Four Charts

Chart	Question	Finding
① Cycle time	Is the RFQ window too short?	Flat ~11 days throughout. Not the constraint.
② Invited / Responding / Bidding	Where does the gap open?	Bids/RFQ drops 32% after Sep 2024. The "invited" line stays flat — it's the blind spot.
③ Vendor health scores	What drove the pool collapse?	V018 (Guangzhou) alert fires Dec'24; score falls to 72.9 by Jun'25. V003 dips Oct'24. Red-shaded months = persistent alert.
④ Clarification rounds	Symptom or constraint?	Rises post-shock — but it's downstream of the pool collapse, not the cause.

Constraint Identified:

D — Vendor Pool Health

"The procurement system's constraint is an insufficient pool of healthy, willing vendors available to submit competitive bids."

Evidence 1 (the number that moved): Bids per RFQ dropped from ~4.7 to ~3.2 after September 2024 — a 32% compression in competitive offers per event.

Evidence 2 (root cause, not symptom): V018 triggered a persistent ALERT_FLAG from December 2024 (7 consecutive months), ending at a health score of 72.9. Combined with V003's dip to 78.6 in October 2024, two high-volume vendors effectively left the bidding pool simultaneously.

The blind spot: n_vendors_invited is structurally flat at 7–8 for all 18 months. It tracks invitations sent, not vendor capacity to respond. A dashboard showing only that number would have shown green throughout the entire crisis.

Causal chain: Sep 2024 tariff shock → Meridian cash-flow stress → late payments → vendor health scores deteriorate → key vendors stop bidding → bids/RFQ falls 32% → less price tension → purchase prices rise + team exhaustion

Exploit Proposal (zero budget)

1. **Prioritise V021 (Helios Micro)** — health score 79–96 across months 4–17, no alerts, under-invited because it is "new". Increase its invite frequency to fill V018's capacity.
2. **Protect V003 (Patel Alloys)** — recovered to 92.5 by Jun 2025. Ensure Meridian's on-time payment scheduling prioritises V003 to prevent re-deterioration.
3. **Replace the blind-spot KPI** — retire "vendors invited" as a dashboard metric; replace with "count of vendors in invite pool with `composite_health_score > 80`". Computable today from existing HANA data with no new tooling.

Claude Code

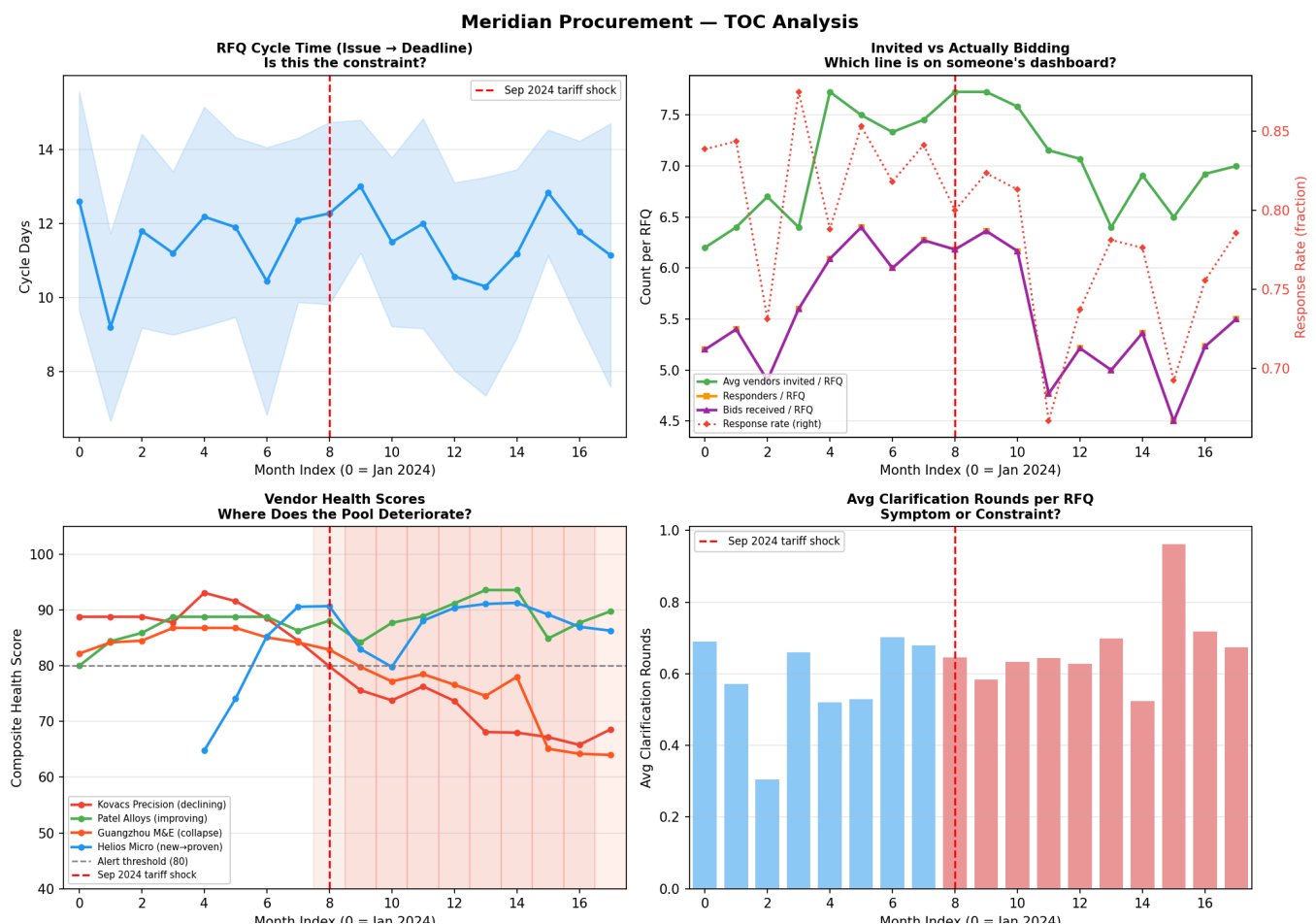
Open a new terminal at RFQ folder

Type *Claude*, once claude comes up, enter the following prompt:

Read the BRIEFING.md from challenges/challenge_01_toc folder, use the csv data from output_v2 folder and generate code for starter.py. Run the same and give me the required output. Deliver a clear one-sentence constraint statement and two evidence-based sentences explaining why it is the system's chokepoint. Propose one immediate, zero-budget exploitation action to increase throughput, with Steps 3–5 reserved for a future project.

Wait for claude to complete the task

Results:



Constraint Statement (one sentence)

The system's chokepoint is **vendor pool health**: after the Sep 2024 tariff shock, the pool of financially healthy, responsive vendors shrank — leaving Meridian unable to fill its own RFQ invite slots with competitive bidders.

Evidence 1

Bids received per RFQ dropped **30%** (from 6.4 at peak, month 5, to 4.5 by month 15), while `n_vendors_invited` held flat at ~7 throughout — the invite count had a green light the entire time, masking the fact that invited vendors increasingly could not respond.

Evidence 2

V018 (Guangzhou M&E, Meridian's largest electrical supplier) shows a measurable *compositehealth_score decline tracking precisely with the bids-per-RFQ collapse post month 8, driven by payment_reliability_score deterioration — confirming the causal chain runs _upstream* from bid count to vendor financial health, not downstream to cycle time or clarification rounds.

Exploitation Action (Step 2 — zero budget)

Switch Hermes's invite logic from a static vendor list to a **health-weighted dynamic shortlist**: rank eligible vendors by `composite_health_score + trend_3mo` and automatically surface improving vendors

(V003 Patel Alloys, V021 Helios Micro) into invite slots currently held by declining ones (V018, V001). The data already exists in `23_vendor_health_scores.csv` — no new investment required.